

34 The L2 Acquisition of Spanish Phonetics and Phonology

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1 Introduction

Perhaps somewhat surprisingly, the acquisition of the phonetics and phonology of Spanish as an L2 has received relatively little attention, much less than other aspects of the grammar (see Montrul, Chapter 35, below). Apparently, applied linguists working on the L2 learning of Spanish have been discouraged by the recurrent view that phonology is “too difficult” for L2 learners and perhaps less important for effective communication than other aspects of the language. For instance, Pennington and Richards (1986: 207, cited in Elliot 1997: 95) claim that “pronunciation, traditionally viewed as a component of linguistic rather than communicative competence or as an aspect of accuracy rather than conversational fluency, has come to be regarded as of limited importance.” As a consequence, relatively little is known about the stages of the L2 acquisition of Spanish sounds, the difficulties that may be caused by certain Spanish sound patterns (and not others) to learners of different linguistic backgrounds, the potential factors (social, instructional, cognitive, anatomical, etc.) that may affect the learning of some sounds (and perhaps not others), etc. Much remains to be investigated. In particular, there is a need for studies that use sophisticated experimental methods and tools available to researchers in other fields since these tools can be more sensitive in revealing the relative relevance of different factors. Finally, systematic comparisons should be performed with language learners differing in native language and linguistic experiences. A need is evident even at the basic level of data acquisition, theorization, and modeling.

This is not to say, of course, that no work has been carried out on the topic that concerns us here. In fact, the study of the L2 acquisition of Spanish sounds has a long tradition. For instance, working within the tenets of Lado’s Contrastive Analysis Hypothesis (CAH) (Lado 1957), early research was carried out which attempted to predict the difficulties native English learners of Spanish would face.

This was done by systematically comparing the phonetics and phonology of English and Spanish (Lado 1956; Bowen and Stockwell 1960; Cárdenas 1960; Stockwell and Bowen 1965). The main hypothesis was that the “deviations in pronunciation” seen in L2 learners are mainly (or solely) due to the specific characteristics of their L1. This early view (Trubetzkoy 1939), which is based on the anecdotal observation that learners with different L1’s have non-native accents with drastically different characteristics, has been incorporated in modified form in (and largely confirmed by) modern, fully-developed theoretical models, such as Best’s Perceptual Assimilation Model (PAM) (Best 1995) and Flege’s Speech Learning Model (Flege 1995). Finally, some recent work has focused on investigating the potential effects of formal instruction or context of learning on the acquisition of pronunciation patterns (e.g., Elliot 1997; Díaz-Campos 2004), thus concentrating on progress or development. The present chapter reviews a few recent experimental studies on the learning of Spanish vowels and consonants. Prior to discussing laboratory data, we proceed to introduce some general aspects of second language phonetic learning.

2 Second-language speech learning and bilingual phonetics

Learning a second language (L2) is a challenging enterprise. One of the aspects of the second language that presents a greater challenge for the learner is the development of speech patterns that resemble those possessed by native speakers of the target L2. Many highly fluent, proficient adult bilinguals display the following: retention of a noticeable non-native accent in their L2 (as judged by monolingual speakers of the L2) even after having had extensive experience with the language (e.g., Oyama 1976; Flege et al. 1995a; Flege et al. 1999a); production of L2 speech sounds that differ from those produced by native speakers of the target L2 (e.g., Caramazza et al. 1973; Bohn and Flege 1992; Flege et al. 1992, 1995b, 1999b; Munro et al. 1995; MacKay et al. 2001; Piske et al. 2002); and demonstration of some difficulties perceiving and processing L2-specific sounds and sound contrasts (e.g., Caramazza et al. 1973; Pallier et al. 1997; Sebastian-Gallés and Soto-Faraco 1999; Bosch et al. 2000; MacKay et al. 2001; Flege and MacKay 2004; Hojen and Flege 2006). Numerous factors have been found to affect the ability of L2 learners to acquire the speech skills of native speakers of the L2. In a review, Piske et al. (2001) discuss the following factors: age of L2 learning, length of residence in an L2-speaking environment (length of experience with the L2), gender, length of formal instruction, motivation, individual language-learning aptitude, and amount of L1 vs. L2 use. The single factor that has received the most attention in the experimental phonetic literature is chronological age at the onset of L2 learning.

It is by now a proven fact that age of learning of the L2 is a significant, robust predictor of perceived foreign accent (e.g., Oyama 1976; Flege et al. 1995a), it affects the acoustic characteristics of sounds produced by L2 speakers (e.g., Caramazza et al. 1973; Guion 2003), and it also affects the perceptual abilities of L2 learners

(e.g., Caramazza et al. 1973; Pallier et al. 1997; Sebastián-Gallés and Soto-Faraco 1999). However, it is unclear what the cause of this might be (Flege 1987a; Long 1990; Flege et al. 1995a). Lenneberg (1967) proposed that the finding was due to the existence of a critical period for language learning (i.e., the Critical Period Hypothesis or CPH). According to this explanation, the complete mastery of a language is no longer possible if the onset of learning occurs after the end of some period in life during which human beings retain a full language-learning capacity. The existence of a critical period tends to be attributed to an age-dependent loss of neural plasticity or to maturational changes in the structure and functioning of the brain that occur during puberty (e.g., Lenneberg 1967; Patkowski 1980, 1990; Scovel 1988; DeKeyser 2000).

An alternative hypothesis to the CPH that has received recent attention is that age effects on L2 speech performance are due to differences in the type of interactions that take place between the L1 and the L2 sounds (e.g., Flege 1995, 2002, 2007; see also Pallier et al. 2003). According to this proposal, individuals who have had a longer experience with their L1 at the time of onset of L2 learning have greater difficulties in learning the speech skills of their L2 because the likelihood that L1 sounds interfere in the learning process of specific L2 sounds is greater. It is known that children become attuned to their native language very early in life (Werker and Tees 1984; Kuhl et al. 1992; Kuhl et al. 2008). On the one hand, very young infants are able to correctly perceive the differences among virtually all sound contrasts (Werker and Tees 1984; Kuhl et al. 2006; see Kuhl et al. 2008). On the other hand, Kuhl et al. (2008) discuss some evidence suggesting that, by the end of the first year of life, the abilities of individuals to discriminate among non-native sound categories diminishes and, importantly, their ability to correctly identify native phonetic units increases. Most surprisingly, children who retain robust perception of non-native phonemic contrasts by the end of their first year of life have been found to show slower subsequent advancement in their native language (Kuhl et al. 2008 and references therein). Therefore, it is not inconceivable that the development of robust native sound categories will affect the development of L2 sound representations and that L1 interference on the L2 will depend on the stage of development of L1 sound categories. Consequently, recent research has pursued the idea that differences between native and non-native speakers are due to “perceptual interference” of one system on the other (e.g., Iverson et al. 2003, 2008).

Two recent theoretical models have offered operationalizations of the interactions that may develop between L1 and L2 sounds: that is, Flege’s Speech Learning Model (SLM) (e.g., Flege 1995) and Best’s Perceptual Assimilation Model (PAM) (e.g., Best 1995). While Best’s theory was designed to explain cross-language speech perception in general (that is, perception of non-native sounds by naive listeners), Flege’s theory was designed to model the *development* of non-native speech skills by L2 learners. In other words, while both models are useful to predict the “problems” non-native listeners may experience, only Flege’s model includes an account of how these “difficulties” might be overcome by experienced learners. Both models make explicit and testable predictions and are backed up by decades of experimental research. Flege’s SLM assumes that the development of speech skills

remains intact across the life span of individuals and that L1 and L2 sound categories reside in a common phonetic space. Therefore, the model assumes that interactions will arise between L1 and L2 sound categories. On the one hand, if there is little acoustic-perceptual distance between an L1 and an L2 sound, learners are likely to “equate” the two sounds (i.e., perceive an L2 sound through the grid of the L1) and thus postulate a single sound category for the two sounds. Although bilinguals might eventually be able to overcome equivalence classification provided sufficient and appropriate input, this phenomenon may be responsible for blocking the development of L2-specific sounds in the case of many individuals or for delaying it in the case of others. On the other hand, the development of “new,” L2-specific sounds is more likely to occur when there are larger acoustic-perceptual differences between the target L2 sound and the closest L1 sound (Flege and Hillenbrand 1984; Flege 1987b). In the event that an L2 learner develops a separate category for an L2-specific sound, the SLM contemplates the possibility that the L1 and the L2 sound categories might deflect away from each other in order to increase contrastiveness between the two sounds (see also Guion 2003).

The development of explicit models of L2 speech learning that look into L1–L2 interactions make it possible to meaningfully address research questions that seem to be basic for our understanding of the L2 acquisition of phonetics and phonology. For instance, since L2 learners may indeed make progress in acquiring the sounds of their L2, how is it that they develop this knowledge? May we explain the stages and processes L2 learners experience in their acquisition of new sounds and articulatory patterns? Are these stages and processes stable across individuals learning the same language? Are there similarities in the experiences of learners trying to acquire different languages? Are there principled reasons why some sounds appear to be “harder” than others in general (e.g., Colantoni and Steele 2008)? How may we address and understand individual differences in L2 speech learning? How may we better train learners to acquire the speech skills of their L2? What constitutes sufficient and appropriate input? Most of these questions have yet to receive a satisfying answer. We now proceed to discuss some experimental Spanish data available to us. We will focus on segmental phenomena.

3 Spanish vowels

Recent research has investigated the production and perception of Spanish vowels by native speakers of several languages, including Dutch (Escudero and Boersma 2002; Iruela 2003), English (Morrison 2003; García de las Bayonas 2004; Cobb 2009; Menke and Face 2010), Japanese (Carranza 2008), Ecuadorian and Peruvian Quechua (Guion 2003; O’Rourke 2010), K’iche’ Maya (Baird 2010) and Mandarin Chinese (Chen 2007), among others. However, the study of the L2 perception of Spanish vowels has attracted very little attention. This is hardly a surprise if we take into consideration that Spanish is a language with a relatively small vowel inventory. Recent evidence shows that learners whose L1 has a larger vowel inventory than their L2 tend to be faster or more successful in developing

native-like L2 vowel perception skills than learners whose L1 has a smaller vowel inventory than their L2 (e.g., Iverson and Evans 2007, 2009). It is expected that L2 learners of Spanish whose native language has a larger vowel inventory will assimilate the five Spanish vowels to five different L1 vowel categories. Therefore, even though it is likely that the Spanish vowel categories of L2 learners will not mirror those of native speakers (at least not in production), it is expected that the learners will experience no difficulty in discriminating the five vowels of Spanish from each other. For instance, Spanish /i e a o u/ may be assimilated to the English vowels in *sea*, *day* or *debt*, *black*, *mow*, and *moon*, respectively. García de las Bayonas (2004, cited in Menke and Face 2010) asked native English-speaking L2 learners of Spanish to identify Spanish vowels. Identification accuracy was very high, between 94% and 96%.

A language with a relatively small vowel inventory may allow for a wider acoustic distribution of each of these vowels than one with a larger vowel inventory. If that occurs, it is possible that the acoustic space occupied by a single L2 vowel overlaps with those of two or more L1 vowels. How does an L2 learner, solely on the basis of positive evidence, learn that there is no phonemic contrast between variants (allophones) of an L2 sound that perceptually correspond to two L1 phonemes? Escudero and Boersma (2002) referred to this theoretical possibility with the term “multiple category assimilation,” following the terminology of Best’s PAM. In the case of multiple-category assimilation, a further source of error could arise if one of the two (or more) L1 vowels linked to a single L2 vowel partially overlaps with a second L2 vowel (Morrison 2003). In the most likely case, however, if a vowel category results from the merger of a single L1 and a single L2 vowel with different acoustic-perceptual distributions (i.e., with a partial overlap between the two or with one being a subset of the other), the production of the L2 learner will deviate from the native speaker norm and his perceptual boundaries may also be dislocated.

Even though significant difficulties in Spanish-vowel discrimination by speakers of languages with a larger vowel inventory are not expected (i.e., “significant” in the sense that they are unlikely to affect lexicalization or lexical access) this does not mean that perceptual differences between natives and non-natives do not arise. Morrison (2003) showed that native English speakers assimilated Spanish /a/ to the English vowel in *hat* at a rate of 70%, to the vowel in *hut* 14% of the time, to the vowel in *hot* 5% of the time and to the vowel in *het* at a rate of 6%. Also, both Spanish /o/ and /e/ were assimilated to English /e/ and /o/ 81% of the time, and 19% of the time to other vowels. On the other hand, Spanish /i/ was assimilated to the English vowel in *heat* at a rate of 100% and Spanish /u/ was assimilated to the English vowel in *hood* 91% of the time. That is, even though English speakers’ discrimination of Spanish vowel contrasts was very good, it was not at ceiling. Additionally, Escudero and Boersma (2002) showed that Dutch-speaking learners of Spanish made significantly more errors when identifying Spanish /i/ and /e/ than when identifying Spanish /o/ and /u/. Further tests by these authors suggested that this was due to the patterns of assimilation of Spanish vowels with native Dutch vowels. Dutch has three front vowels that virtually occupy the acoustic-perceptual space of

the two Spanish front vowels (three-to-two correspondence) while it has two back vowels that virtually occupy the space of the two Spanish back vowels (two-to-two correspondence). Escudero and Boersma's interpretation is that the existence of more contrasts in the L1 than the L2 of these learners in, approximately, the same acoustic-perceptual region affects the formation of perceptual category boundaries. These two papers show that knowledge of the acoustic and perceptual (phonetic) characteristics of both the L1 and the L2, in addition to the system of categorical contrasts (phonology), is fundamental in order to make robust hypotheses about the perceptual development of L2 sounds. This suggests that purely symbolic (phonological) learning models should be abandoned in favor substance-based (phonetic) models.

We now turn to vowel production. Guion (2003) carried out a thorough acoustic study of the Spanish and Quichua (Ecuadorian Quechua) vowels produced by four groups of Quichua-Spanish bilinguals residing in Otavalo. These four bilingual groups differed in their self-reported age of onset of learning (AOL) of Spanish: (1) simultaneous bilinguals (AOL = 0); (2) early bilinguals (AOL = 5–7); (3) mid bilinguals (AOL = 9–13); and (4) late bilinguals (AOL = 15–25). Quichua has a system of three vowels /i a u/. Spanish /a/ is lower (i.e., has a higher F1) than Quichua /a/. Quichua /i/ falls somewhere in between the acoustic (F1–F2) distribution of Spanish /i/ and /e/, with more overlap with Spanish /i/ than /e/. Quichua /u/ is acoustically located somewhere between the acoustic distribution of Spanish /u/ and /o/.

In addition to contrasting the different speaker groups to investigate potential AOL effects, Guion (2003) examined the potential L1–L2 interactions within each speaker's vowel systems. Regarding the production of front vowels, the results revealed that bilinguals could be classified into three groups: (1) speakers who had only one phonetic category for their Quichua /i/ and both their Spanish /i/ and /e/ (one-vowel group); (2) speakers who assimilated their Quichua /i/ to their Spanish /i/ but maintained a separate category for their Spanish /e/ (two-vowel group); and (3) speakers who developed new categories for both their Spanish /i/ and /e/ and thus maintained a separate category for their Quichua /i/ (three-vowel group). "Success" in new-category formation was indeed directly affected by AOL, with most simultaneous bilinguals being able to maintain three separate front vowel categories and most late bilinguals presenting a system with a single front vowel. Regarding the back vowels, also three groups were found: (1) speakers who used fundamentally the same vowel for their Quichua /u/ and their Spanish /u/ and /o/ (one-vowel group); (2) speakers who assimilated their Spanish /o/ to their Quichua /u/ but kept a separate phonetic category for their Spanish /u/ (separate Spanish /u/); and (3) speakers who assimilated their Spanish /u/ with their Quichua /u/ but maintained a difference with their Spanish /o/ (separate Spanish /u/). Interestingly, none of the twenty bilinguals was able to develop a three back-vowel system. Finally, Guion (2003) found that the bilinguals who had been able to develop new categories for either their Spanish front *or* back vowels also presented two separate statistical distributions for their Quichua /a/ and their Spanish /a/.

In summary, Guion (2003) infers that late learners tend to use Quichua vowel categories in their Spanish, unlike early and simultaneous bilinguals, who seem to have been able to develop L2 categories that resemble those used by Spanish-speaking monolinguals. Guion (2003) is a fundamental study in that it provides a full acoustic characterization of both the L1 and L2 vowel systems of a large controlled group of bilinguals and late L2 learners. This robust analysis, with its many comparisons, was able to reveal differences across speakers as well as the nature of phonetic–phonological systems in bilinguals. In that sense, this article stands as a capital study of the L2 production of Spanish vowels.

While Quichua-speaking learners of Spanish need to develop a system with *more* vowels than the one they possess in their L1, the situation is the opposite for native speakers of languages such as English, which has more vowel categories than Spanish. We have already considered that these two situations pose different kinds of perceptual learning difficulties for L2 learners. Thus, while Quichua natives will have to learn new contrasts and thus develop new categories for Spanish, English natives will have to learn the appropriate acoustic (F1–F2) boundaries of Spanish vowels although they may not have to develop a completely new phonetic category for most of them. Menke and Face (2010) and Cobb (2009) (see also Cobb and Simonet 2010) examine the acoustic characteristics (F1–F2) of Spanish vowels as produced by native English speakers differing in length of experience with Spanish and, thus, level of proficiency in the language. Perhaps the most important conclusion of both Menke and Face (2010) and Cobb (2009) is that, while native English-speaking learners differ greatly in their production of Spanish vowels from native Spanish-speaking controls at the initial state, they tend to improve with increased experience with the language. Interestingly, the English-speaking learners display different patterns of development or “progress” as a function of the different Spanish vowel categories.

Although there are some interesting similarities between the Menke and Face (2010) and Cobb (2009) studies, there are also some differences. Both studies confirmed that Spanish /a/, /u/ and /e/ present some initial problems to native English learners. While beginners’ /u/ tokens are maximally different from those of native controls (apparently more so than any other vowel), advanced learners do not differ, for the most part, from native speakers in this respect. On the other hand, the differences between these two studies are somewhat puzzling. While Menke and Face (2010) found significant differences between learners and natives in their production of /i/ and /o/, Cobb (2009) failed to find these. The /a/ tokens produced by all three groups of learners in Menke and Face (2010) differed from the native controls, suggesting that the pronunciation of /a/ is affected even at a very advanced stage. However, Cobb (2009) did not find significant differences between advanced learners’ and natives’ /a/, while they were found between beginners and advanced learners. Finally, while Cobb (2009) did not find evidence for pronunciation “improvement” in the case of /e/ with even advanced learners differing from native controls, Menke and Face (2010) did. Therefore, the pattern of /a/ and /e/ improvement was reversed between the two studies. It is unclear why these differences arose, which calls for more research on this specific topic. One aspect to

keep in mind is that the two studies introduce different types of noise (or error) in their data. On the one hand, while Cobb (2009) uses only five speakers per group, Menke and Face (2001) use as many as twenty. On the other hand, while Cobb (2009) carries out a careful normalization of formant values (see Cobb and Simonet 2010 for a third normalization procedure), Menke and Face (2010) pool data from many speakers without prior normalization.

A second aspect investigated in both Menke and Face (2010) and Cobb (2009) is unstressed vowel production. Menke and Face (2010) compared tokens of each of the five stressed vowels with their corresponding counterparts in unstressed position. The analysis was limited to establishing how many (and which) of the vowels would show an effect of lexical stress. Stress effects were found for the native controls in the F2 of /e/ and /u/, but not of /i/ or /o/. In the beginners' productions, effects were found for F2 in /e/ and /u/ and for both F1 and F2 in /a/, but not in /i/ and /o/. Significant effects were revealed for the intermediate learners' /a/ and /i/ in both F1 and F2 and in the F2 of /e/ and /u/, but not in /o/. Finally, the advanced learners displayed stress effects for all of the vowels (i.e., F1 and F2 in /a/, F1 in /o/ and F2 in /i/ and /u/). This finding is unexpected. Greater stress effects were found for the advanced learners than for the beginners with the intermediate learners located in the middle. Paradoxically, while the stressed vowels of native English speakers tend to get "better" with improvement in proficiency, their unstressed vowels tend to get worse.

Cobb (2009) investigated this topic from a different perspective. She calculated the centroid F1 and F2 for the entire vowel system of each individual participant and then measured the Euclidean distance between all the vowels, stressed and unstressed, produced by a given participant and their own centroid. The findings were also unexpected. Native controls were found to centralize more than the two learner groups and the advanced learners were found to centralize more than the beginners. The paradox revealed by Menke's and Face's (2010) findings is not evident in Cobb's (2009) data; that is, learners' productions of *both* stressed and unstressed vowels "improve" (i.e., approach native speakers' productions) with an improvement in the overall proficiency level. An important difference is that Cobb (2009) found robust effects of stress in the natives' vowel productions while Menke and Face (2010) did not.

Cobb and Simonet (2010) reanalyzed Cobb's data by investigating the effects of stress on both the F1 and F2 of all vowels. Effects of stress were found for all groups and all vowels, but these effects differed as a function of both factors. It was found that the three groups centralized their /i/ and /u/ vowels in unstressed positions and that this effect was of a similar size across groups. However, there were significant differences in the mid vowels. For /e/, native speakers were found to modify F2 (centralize), while beginners were found to modify only F1 (raise). For /o/, natives were also found to modify F2 (centralize) while the two groups of learners were found not to modify the acoustic characteristics of their unstressed mid-back vowel as a function of stress.

To summarize, acquiring language-specific patterns of conditioned phonetic variation is apparently "difficult." While English-speaking learners may make

rapid progress in their pronunciation of stressed vowels in optimal positions, acquiring vowel-specific patterns of stress reduction takes more time. Perhaps it is now the appropriate time to address the study of the acquisition of L2-specific conditioned phonetic variation and move beyond the acquisition of L2-specific sound contrasts in salient prosodic positions. Importantly, the different conclusions reached by the three studies suggest that findings largely depend on the analytical procedures or tools we decide to use to examine our data. Future studies would benefit from more advanced and reliable acoustic and statistical analyses and a more sensitive placement of subjects into groups.

4 Spanish consonants

A series of studies have investigated the production of Spanish consonants by non-native speakers. We will focus on a number of recent articles that have investigated the production and perception of the following Spanish sounds: (1) voiceless and voiced stops produced by early Spanish–English bilinguals and late English-speaking L2 learners of Spanish (e.g., Zampini 1998; Magloire and Green 1999; Zampini and Green 2001); (2) intervocalic voiced stops/approximants produced by late English-speaking L2 learners of Spanish (e.g., Zampini 1994; González-Bueno 1995; Díaz-Campos 2004; Face and Menke 2009); (3) the intervocalic rhotic contrast between the tap and the trill produced by English-speaking adult learners of Spanish (e.g., Major 1986; Reeder 1998; Waltmunson 2005; Face 2006; Rose 2010); and (4) alveolar laterals produced by early Catalan–Spanish bilinguals (Simonet 2010a, 2010b).

Perhaps the feature of the sound system of Spanish that has received more attention from the L2 speech learning literature is the production and perceptual discrimination of the contrast between voiceless and voiced stops (e.g., /p/–/b/; see Zampini and Green 2001 for a review). Spanish /p/ differs from /b/ (when realized as a stop) in at least two phonetic features: (1) voice onset time (VOT) pattern and (2) length of stop closure. Voice onset time (VOT) pattern refers to differences in the relative synchronization of the release of the articulators after closure and the onset of voicing (e.g., Lisker and Abramson 1964). Spanish /p/ has a short-lag positive VOT (i.e., the onset of voicing occurs slightly after the time of release of the articulators), while Spanish /b/ has a negative VOT (i.e., the onset of voicing occurs during closure, that is, before the release of the articulators). Additionally, Spanish /b/ (when realized as a stop) presents a significantly shorter closure than Spanish /p/ (Martínez-Celdrán 1993). English /p/ differs from /b/ in that the latter has a short-lag positive VOT while the former has a long-lag positive VOT. There are no differences in the length of the stop closure in word-initial position between these two sounds (e.g., Lisker and Abramson 1964; Zampini 1998; Zampini and Green 2001).

Zampini (1998) (see Zampini and Green 2001) studied the production and perception of the English and Spanish /p/–/b/ contrast by adult L2 learners of Spanish. These subjects were examined four times, three in Spanish (week 3,

week 6, and week 15 of an academic semester) and one in English. The subjects were found to produce Spanish/p/ with a much shorter VOT than English/p/ and a slightly longer VOT than English/b/. Spanish/b/ was produced with a short-lag positive VOT. Importantly, the VOT of Spanish/b/ did not differ from that of English/b/. In sum, these subjects had developed a new category for Spanish/p/ but, apparently, not for Spanish/b/. Spanish/p/ and /b/ were found to be significantly different in their VOTs but this difference was of a much smaller magnitude than that between English/p/ and /b/. The results for length of stop closure revealed that learners produced Spanish/p/ with a much longer closure than Spanish/b/ while English/p/ and /b/ did not differ in this acoustic feature. These results suggested that the L2 learners had been able to develop new categories for *both* Spanish/p/ and /b/ and were successfully marking the contrast in their production; however, Spanish/p/ resembled that of Spanish natives more than Spanish/b/ did. Apparently, thus, Spanish/b/ is an especially “difficult” sound for native English speakers (i.e., although they are able to develop ways of producing the contrast, it is not the case that they do so in a native-like manner).

Zampini (1998) (see Zampini and Green 2001) also examined the auditory discrimination of the Spanish/p/–/b/ contrast of these L2 learners. This was done by investigating the /p/–/b/ VOT perceptual boundary; that is, the VOT value that triggers a higher percentage of selection of one category than the other in a perceptual identification experiment in which a continuum of /p/–/b/ tokens is created by modifying VOT values. In addition to the L2 learners of Spanish tested at three different times, Zampini (1998) recruited English monolinguals and early Spanish–English bilinguals. The results showed that the L2 learners did not differ from the English monolinguals in the first session while they did in the second and third sessions. The learners did not differ from the early bilinguals in the second and third sessions. In other words, evidence of development (i.e., a reorganization of the perceptual abilities of late L2 learners) was found. The significance and completeness of Zampini’s (1998) study is twofold: (1) it investigates the production and perception, and the relationship between the two, of both English and Spanish contrasts as produced by late L2 learners of Spanish, early Spanish–English bilinguals and English-speaking monolinguals; and (2) it “provides longitudinal data, not often seen in the literature” (Zampini and Green 2001: 38). For these reasons, Zampini’s research stands as a very significant step towards our understanding of the L2 acquisition of Spanish stop consonants.

Spanish/b d g/ are realized as approximants or spirants in most contexts, including the intervocalic position. At least four recent papers have investigated the production of these sounds by native English-speaking L2 learners of Spanish (see above). Since English/b d g/ are not often realized as approximants, there is an initial tendency for English-to-Spanish transfer. Zampini (1994) found that a group of late English-speaking L2 learners produced many more stops than spirants in the contexts where native Spanish speakers almost categorically produced spirantized variants. Another difference between the two groups of speakers was that learners

produced some /b/ tokens as labiodental fricatives instead of bilabials. Zampini attributes this difference to the interference of Spanish orthography since the /b/ tokens that received a labiodental realization seemed to be those written with *v* in the standard orthography (see Face and Menke 2009 for further tests on the effects of orthography). Another interesting result of Zampini (1994) is that the learners produced less spirantized variants (i.e., more “errors”) of /d/ than of /b/ or /g/. Zampini attributed this finding to the phonemic status of the English voiced interdental fricative, which is phonetically very similar to the spirantized realization of Spanish /d/. Thus, apparently, learners were less likely to produce approximant /d/ than /b/ or /g/ because they have a native phonemic contrast between two sounds that are allophonic in Spanish. Therefore, they might have been less ready to accept the interdental approximant as an allophonic variant of /d/. Similar results were found in later research. Thus, even though González-Bueno (1995) found an overall higher percentage of spirant realizations in the speech of late L2 learners than Zampini (1994) and also Díaz-Campos (2004), one finding was very similar: /d/ was the phoneme that showed the lower percentage of spirant productions.

Face and Menke (2009) investigated the productions of a large number of English-speaking L2 learners as a function of their L2 proficiency (i.e., advanced, intermediate, beginners). This is the only paper on L2 /b d g/ production to include advanced learners. The results showed cross-sectional evidence of “progress” in production since advanced learners produced more approximants than intermediate learners and intermediate learners produced more approximants than beginners. The rates of approximant realizations were much higher across groups than those in Zampini (1994) and Díaz-Campos (2004) but similar to those in González-Bueno (1995). However, one finding was identical: learners produced less spirant realizations of /d/ than of /b/ or /g/. The finding that the development of spirant realizations of /d/ is somewhat more “difficult” than that of /b/ and /g/ for native English L2 learners of Spanish seems therefore to be a robust one. In light of these findings, Zampini’s (1994) original hypothesis deserves to be submitted to further empirical scrutiny.

The remainder of this section focuses on the L2 acquisition of liquids (rhotics, laterals and stop + liquid clusters) for which mostly production data are available. Face (2006) studied the production of the Spanish intervocalic rhotic contrast as produced by English-speaking late L2 learners of Spanish (see also Major 1986; Reeder 1998; WalTMunson 2005; Rose 2010). Forty-one native English speakers who were learning Spanish in a formal setting were recruited and divided in two groups (i.e., beginners and intermediate learners). Face quantified the type of “errors” the learners made and classified these in two groups: (1) transfer errors, in which learners produce an English-like approximant instead of a Spanish-like tap or trill; and (2) developmental errors, in which learners produce a sound other than the expected Spanish-like tap, the Spanish-like trill or the English-like approximant. First, an important finding was that English-speaking L2 learners of Spanish are generally able to acquire a native-like pronunciation of the Spanish tap at a relatively early stage in their

development. Face attributes this to the fact that there exists a very similar phonetic category in English. Second, regarding the production of the trill, beginners produced native-like trills in only 5% of the instances and intermediate learners produced them in only 26.6% of the instances. Interestingly, the beginners produced as many English-like approximants as other sounds, including taps. On the other hand, the intermediate learners produced many more taps than English-like approximants. Notice that the overall number of Spanish-like (i.e., “correct”) taps is much higher overall than that of trills. Thus, cross-sectional evidence of “progress” was evident in the acquisition of both Spanish rhotics. While the beginners produced large numbers of English-like approximants for both the Spanish tap and the trill, the intermediate learners were mostly “successful” in avoiding English-like approximants in their Spanish production. However, the intermediate learners were not at all successful in acquiring the Spanish rhotic *contrast* since they seemed to overgeneralize the tap in their production of both Spanish rhotic categories.

Simonet (2010a, 2010b) investigated the production of alveolar laterals in two groups of early Spanish–Catalan bilinguals residing on Majorca. Catalan laterals are known to be velarized while Spanish ones are known to be nonvelarized. Ten Catalan-dominant bilinguals were found to produce Spanish laterals with a lower F2 (more velarized) than ten Spanish-dominant speakers. Simonet (2010b) also examined the within-speaker patterns of interaction between L1 and L2 categories and found that the Catalan-dominant males and females, as well as the Spanish-dominant males, but crucially not the Spanish-dominant females, produced Catalan and Spanish laterals with different acoustic distributions. In other words, while most bilinguals produced two different phonetic categories in their two languages, some did not. Simonet attributes this finding to the effect of the social indexicality of the Catalan-like velarized /l/, which seems to be a sociophonetic stereotype in Mallorca. Simonet’s main finding was that social indexical constraints may greatly affect bilingual speech performance, perhaps as much as cognitive factors do.

Colantoni and Steele (2006) examined the production of Spanish stop + liquid clusters such as /pr/, /br/, /pl/, /bl/, etc. Ten native English-speaking L2 learners of Spanish were recruited and classified in two groups (i.e., intermediate and advanced learners). The authors tested very specific hypotheses regarding the stages of development of native-like pronunciation of the different acoustic characteristics of the Spanish stop + liquid clusters. Learners differed from native Spanish speakers in most of the features of the target sound sequences, but especially in their production of the rhotics and in their use of epenthetic vowels. Finally, an interesting hypothesis that received support in these data was that L2 learners were more successful in acquiring L2 features that possessed a smaller degree of variability in native speech while sounds or features that were highly variable in native productions were “difficult” to learn. Future research on the L2 acquisition of these or other sound categories will do well to investigate the magnitude of variability of the target sounds in the speech of native speakers prior to analyzing L2 productions.

5 Conclusions

The present chapter has reviewed some of the recent experimental work on the L2 acquisition of Spanish phonetics and phonology. We have focused on papers that studied the development of native-like production and perception of Spanish segments in early and late bilinguals or L2 learners. Regarding Spanish vowels, we have discussed research on (1) the production of these segments by native Quichua-speaking simultaneous, early and late bilinguals (e.g., Guion 2003), and by native English-speaking late L2 learners (e.g., Cobb 2009; Menke and Face 2010), and also on (2) the perception of vowels by native speakers of English (Morrison 2003; García de las Bayonas 2004) and Dutch (Escudero and Boersma 2002). Regarding Spanish consonants, we have discussed research on (1) the production and perception of voiced and voiceless stops by native English-speaking late learners and by Spanish–English bilinguals (see Zampini and Green 2001), (2) the variable production of spirantized /b d g/ by late English-speaking learners (e.g., Zampini 1994; Face and Menke 2009), and (3) the production of liquids and stop + liquid sequences in late English-speaking learners (e.g., Face 2006; Colantoni and Steele 2009) and Catalan–Spanish early bilinguals (Simonet 2010a, 2010b). The acquisition of many more Spanish sound categories should be investigated. Different native languages should also be examined. It is to be expected that surprising findings will be revealed by an improvement in both the depth and the breadth of our research questions, an adoption of fully developed theoretical models such as Flege’s SLM and Best’s PAM, and a careful use of sophisticated laboratory techniques.

Even though we have decided to focus our attention on the L2 acquisition of Spanish segments, this is not to say that prosodic features have not been investigated. In fact, a number of authors have examined the production of some Spanish prosodic features by both early and late L2 learners, for example, native Mandarin-speaking late L2 learners (Chen 2007), early Basque–Spanish bilinguals (Elordieta and Calleja 2005), simultaneous and early Quechua–Spanish bilinguals (O’Rourke 2005, 2008), and early Catalan–Spanish bilinguals (Simonet 2011). However, while the study of the L2 acquisition of segments presents the advantage of being able to rely on very robust theoretical models, such as Flege’s SLM (e.g., Flege 1995), which allow for the operationalization of specific hypotheses, research on the acquisition of prosodic features is still in need of a fully-fledged, well-grounded model. For instance, while there is no *a priori* reason to assume that the acquisition of prosodic categories will be different from that of segmental categories, it is actually still unclear how to identify what the conceptual equivalent of a segmental phonetic category might be in prosody (see e.g., Flege 1995 for the concept of “phonetic category”). Research on all these fronts is necessary for our understanding of the acquisition of L2 Spanish sounds and of the factors that affect bilingual speech production and perception.

Robust theoretical models are now available that make specific, testable predictions for the behavior of different types of bilinguals and L2 learners. These theories rely on complete cognitive models. Bridges have been built between

findings in L2 speech learning research and findings in fields such as L1 acquisition, sociophonetics, speech perception, and psycholinguistics. Sensitive and reliable methods have been developed in order to be able to obtain a better understanding of speech performance in bilinguals and L2 learners. Researchers working on the L2 acquisition of Spanish phonetics and phonology and/or on speech performance in bilinguals who have Spanish as one of their languages (either as a dominant or a nondominant language) will do well to ground their work on such theories and models and to use the reliable methods that have been developed recently.

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